2 Optimal Ordering

Let $x_i \in \mathcal{X}$ and $y_i \in \{-1, +1\} \quad 1 \le i \le n$. Define

$$\begin{aligned} I_- &= \{i | y_i = -1\} \text{ with } n_- = \text{num. of elements in } I_- \\ I_+ &= \{i | y_i = +1\} \text{ with } n_+ = \text{num. of elements in } I_+ \end{aligned}$$

And let $n = n_- + n_+$. We seek an f which minimizes the functional

$$J(f) = \frac{1}{n_- n_+} \sum_{i \in I_-} \sum_{j \in I_+} (1 - f(f(x_j) - f(x_i))) + \lambda \|f\|_H^2$$

1. Justify that such an f^* can be expressed a function indexed by α of the form

$$f^*_\alpha(x) = \sum_{i=1}^{n} \alpha_i K(x, x_i)$$

Let H as a Hilbert space and let $f \in H$. Since H is a Hilbert space it can be expressed as

$$H = H_\mathcal{V} \oplus H_\perp$$

where both $H_\mathcal{V}$ and $H_\perp$ are orthogonal subspaces. This means any $f \in H$ can expressed as

$$f^* = f_\mathcal{V} + f_\perp.$$

We contruct another subspace of H defined as

$$H_\mathcal{V} = \{f : f(x) = \sum \alpha_i K(x, x_i)\}$$

By construction $K(\cdot, x_i) \in H_\mathcal{V}$, therefore for any $1 \le i \le n$ we have

$$f(x_i) = \langle f_\mathcal{V} + f_\perp, K(\cdot, x_i) \rangle = \langle f_\mathcal{V}, K(\cdot, x_i) \rangle + \langle f_\perp, K(\cdot, x_i) \rangle = f_\mathcal{V}(x_i).$$

Therefore since $f^* \in H$ is the f that minimizes $J(f)$, then we have

$$J(f^*) - J(f_\mathcal{V}) = \lambda(||f_\mathcal{V}||^2 + ||f_\perp||^2) - \lambda||f_\mathcal{V}||^2 = \lambda||f_\mathcal{V}||^2 \ge 0$$

Other the other hand f^* minimizes $J(f)$ and so

$$J(f^*) - J(f_\mathcal{V}) \le 0$$

Both equations together implies $||f_\perp|| = 0$ which futher implies that $f_\perp = 0$.

Hence any $f^* \notin H_\perp$ and so $f^* = f_\mathcal{V}$. I.e all f which minimize $J(f)$ must be in $H_\mathcal{V}$

2. Write J(f) in terms of α.

Substituting the form of $f \in \mathcal{V}$ into $J(f)$ gives

$$J(f) = \frac{1}{n_- n_+} \sum_{ij} \left(1 - \sum_k \alpha_k K(x_k, x_j) - \sum_k \alpha_k K(x_k, x_i) \right) + \lambda \alpha^\top K \alpha =$$

$\frac{1}{n_-n_+}\sum_{ij}\left(1-\sum_k \alpha_k([K]_{kj}-[K]_{ki})\right)+\lambda\alpha^\top K\alpha = \frac{1}{n_-n_+}\sum_{ij}\left(1-\alpha^\top(K_j-K_i)\right)+\lambda\alpha^\top K\alpha =$
$1-\frac{\alpha^\top}{n_-n_+}\sum_{j\in I_+}\sum_{i\in I_-}(K_j-K_i)+\lambda\alpha^\top K\alpha$

with K_j is defined as jth collumn of K and likewise for K_i.
3. Furthermore if we define $K_- = \frac{1}{n_-}\sum_{i\in I_-} K_i$ and likewise, $K_+- = \frac{1}{n_+}\sum_{j\in I_+} K_j$.
Substituting,

$J(f) = 1-\frac{\alpha^\top}{n_-n_+}\sum_{j\in I_+}\sum_{i\in I_-}(K_j-K_i)+\lambda\alpha^\top K\alpha = 1-\alpha^\top\left(\frac{K_+}{n_-}-\frac{K_-}{n_+}\right)+\lambda\alpha^\top K\alpha$

*From discussion in class, that the $n\pm$ in the denominators vanish after summing over i, j.

4. Calculate $\nabla_\alpha J(\alpha)$.

Using $\nabla_\alpha \alpha^\top b = b$ and $\nabla_\alpha^\top K\alpha = (K^\top + K)\alpha = 2K\alpha$ gives

$$\nabla_\alpha J(f) = \left(\frac{K_+}{n_-}-\frac{K_-}{n_+}\right)+2K\lambda\alpha$$

5. Setting the gradient to zero immediately gives α^*:

$$\alpha^* = \frac{K^{-1}}{2\lambda}\left(\frac{K_+}{n_-}-\frac{K_-}{n_+}\right)$$

6. Using the kernel, $K(x,x') = x^\top x'$ with associated RKHS

$$H = f_v; f_v(x) = v^\top x, \text{ with } v\in\mathbb{R}^d \text{ and } \langle f_v, f_w\rangle = v^\top w$$

verify the reproducing property of K.

By inspection, $K(x,\cdot) = f_x \implies$
$\langle f_v, K(x,\cdot)\rangle = \langle f_v, f_x\rangle = v^\top x = f_v(x)$

7. Compute $J(f(\alpha^*))$.

Let $f(\alpha^*) \equiv f^*(x) = \sum_k^n \alpha_k K(x,x_k) = \sum_k^n \alpha_k^* x^\top x_k =$
$\sum_k^n \alpha_k^* x_k^\top x = \beta^\top x$ with $\beta^\top \equiv \sum_k \alpha_k^* x_k^\top$

Hence

$\beta^\top = \sum_k \left[\frac{K^{-1}}{2\lambda}\left(\frac{K_+}{n_-}-\frac{K_-}{n_+}\right)\right]_k x_k^\top =$
$\sum_k \left[\frac{K^{-1}}{2\lambda n_+ n_-}\left(\sum_{j\in I_-} K_j - \sum_{i\in I_+} K_i\right)\right]_k x_k^\top$

Defining $x^+ \equiv \frac{1}{n_+} \sum_j^{n_+} x_j$ and $x^- \equiv \frac{1}{n_-} \sum_i^{n_-} x_i$ gives
$\sum\limits_{j \in I_+} K_j =$

$$\begin{bmatrix} x_1^\top x_{j_1} \\ x_2^\top x_{j_1} \\ \vdots \\ x_n^\top x_{j_1} \end{bmatrix} + \cdots + \begin{bmatrix} x_1^\top x_{j_{n_+}} \\ x_2^\top x_{j_{n_+}} \\ \vdots \\ x_n^\top x_{j_{n_+}} \end{bmatrix} = \begin{bmatrix} x_1^\top \sum\limits_{j \in I_+}^{n_+} x_j \\ \vdots \\ x_n^\top \sum\limits_{j \in I_+}^{n_+} x_j \end{bmatrix} = \begin{bmatrix} x_1^{(1)} & x_1^{(2)} & \cdots & x_1^{(d)} \\ \vdots & \vdots & & \vdots \\ x_n^{(1)} & x_n^{(2)} & \cdots & x_n^{(d)} \end{bmatrix} \left(\frac{x^+}{n_-} \right)$$

and likewise for $\sum\limits_{i \in I+} K_i$ and therefore $\beta^\top$ can written as

$$\beta^\top = \sum_k \left[\frac{K^{-1}}{2\lambda} \left(\frac{K_+}{n_-} - \frac{K_-}{n_+} \right) \right]_k x_k^\top =$$

$$\sum_k^n \left[\frac{K^{-1}}{2\lambda} \begin{bmatrix} x_1^{(1)} & x_1^{(2)} & \cdots & x_1^{(d)} \\ \vdots & \vdots & & \vdots \\ x_n^{(1)} & x_n^{(2)} & \cdots & x_n^{(d)} \end{bmatrix} \left(\frac{x_+}{n_-} - \frac{x_-}{n_+} \right) \right]_k x_k^\top$$

To summerize the method to find $\beta^\top$:

First, caluclate x_+, and x_- from the data points to find $\left(\frac{x_+}{n_-} - \frac{x_-}{n_+} \right)$. This is $(d \times 1)$ vector.

Second, create matrix of data points with componenets in each row. This is a $(n \times d)$ matrix.

Third, calulate the inverse of matrix K. This is an $(n \times n)$ matrix.

Fourth, multiply all of them to produce an $(n \times 1)$ vector.

Use forumula to sum the product of each component with each data point to produce a new $(d \times 1)$ vector.

The optimal f(x) is given by $f^*(x) = \beta^\top x$ with $\beta^\top$ given above.